

1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint_dir, get_model_dir
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4)
os.environ['COLUMNS'] = '100'

# Configuración de reproducibilidad
torch.manual_seed(42)
np.random.seed(42)

# Device
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
```

```
# Print config con formato profesional
config_df = pd.DataFrame({
    'Configuración': ['Device', 'Seed', 'Output Width'],
    'Valor': [str(device), '42', '100']
})

print('\n' + '='*50)
print(' CONFIGURACIÓN INICIAL')
print('='*50)
print(config_df.to_string(index=False))
print('='*50)
```

```
=====
CONFIGURACIÓN INICIAL
=====
Configuración Valor
      Device  cuda
      Seed    42
Output Width  100
=====
```

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'l1-decay',           # L1 con penalty annealing
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 1000,                  # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt', # Path base para cl
    'experiment_base_path': os.path.abspath('../../experiments') ,      # sae/experiments,
    'extras': {
        'expansion': 32,
        'sparsity_coef': 1e-2,
        'epochs': 250,
        'patience': 20,
        'learning_rate': 1e-3,
        'sparsity_decay': 0.98,
```

```

        'batch_size': 256
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')

```

```

Metadata de naming:
variante: standard
tecnica: l1-decay
capa: 6
juegos: 1000
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 32, 'sparsity_coef': 0.01, 'epochs': 250, 'patience': 20, 'learning_1

```

```

extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
) if NAMING_META.get('extras') else None

print(f'extras_id: {extras_id}')

```

```

extras_id: ex5

```

3. Configuración del SAE

Las activaciones ya fueron extraídas previamente con `sae/activations/run_extraction.py`. Aquí especificamos qué archivo cargar y los hiperparámetros del SAE.

```

# Configuración del SAE
CONFIG = {
    # Arquitectura
    'input_dim': 512,
    'hidden_dim': 16384,
    'tied_weights': False,

    # Entrenamiento
    'batch_size': 256,
    'num_epochs': 250,
    'learning_rate': 1e-3,
    'sparsity_coef': 1e-2,
    'sparsity_decay': 0.98,

    # Early Stopping
    'early_stopping': True,
    'patience': 20,

    # Datos
    'activations_file': Path(f"../../activations/data/layer{NAMING_META['capa']}_{NAMING_I"},
    'save_dir': Path('./saved_model'),
    'plots_dir': get_plots_dir(
        NAMING_META['experiment_base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos'],
        extras_id=extras_id
    )
}

# Crear directorio local del modelo (plots_dir ya fue creado por get_plots_dir)
CONFIG['save_dir'].mkdir(parents=True, exist_ok=True)

# Mostrar configuración con formato profesional
config_data = {
    'Categoría': [],
    'Parámetro': [],
    'Valor': []
}

```

```

# Arquitectura
for key in ['input_dim', 'hidden_dim', 'tied_weights']:
    config_data['Categoría'].append('Arquitectura')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Entrenamiento
for key in ['batch_size', 'num_epochs', 'learning_rate', 'sparsity_coef', 'sparsity_decay']:
    config_data['Categoría'].append('Entrenamiento')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Early Stopping
for key in ['early_stopping', 'patience']:
    config_data['Categoría'].append('Early Stopping')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

config_table = pd.DataFrame(config_data)

print('\n' + '='*70)
print(' CONFIGURACIÓN DEL SAE')
print('='*70)
print(config_table.to_string(index=False))
print('='*70)

print(f'\nArchivo de activaciones: {CONFIG["activations_file"]}')
print(f'Expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f'Plots dir: {CONFIG["plots_dir"]}')
print('='*70)

```

```

=====
CONFIGURACIÓN DEL SAE
=====

```

Categoría	Parámetro	Valor
Arquitectura	input_dim	512
Arquitectura	hidden_dim	16384
Arquitectura	tied_weights	False
Entrenamiento	batch_size	256
Entrenamiento	num_epochs	250
Entrenamiento	learning_rate	0.001

```

Entrenamiento sparsity_coef 0.01
Entrenamiento sparsity_decay 0.98
Early Stopping early_stopping True
Early Stopping patience 20
=====

Archivo de activaciones: ..\..\..\activations\data\layer6_1000games.npy
Expansión: 32.0x
Plots dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1000games\ex5\plots
=====

```

4. Dataset de Activaciones (Train/Validation Split 80-20%)

```

class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""

    def __init__(self, activations_path):
        self.activations = np.load(activations_path)

    def __len__(self):
        return len(self.activations)

    def __getitem__(self, idx):
        return torch.tensor(self.activations[idx], dtype=torch.float32)

# Cargar dataset completo
full_dataset = ActivationsDataset(CONFIG['activations_file'])

# Información del dataset con formato profesional
dataset_info = pd.DataFrame({
    'Propiedad': ['Total muestras', 'Shape', 'Dtype', 'Tamaño en memoria'],
    'Valor': [
        f"{len(full_dataset):,}",
        str(full_dataset.activations.shape),
        str(full_dataset.activations.dtype),
        f"{full_dataset.activations.nbytes / 1024**2:.2f} MB"
    ]
})

print('\n' + '='*50)
print(' DATASET DE ACTIVACIONES')

```

```

print('='*50)
print(dataset_info.to_string(index=False))
print('='*50)

# Split train/validation (80-20)
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset,
    [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

# Resumen del split con formato profesional
split_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ],
    'Porcentaje': [
        f"{len(train_dataset)/len(full_dataset):.1%}",
        f"{len(val_dataset)/len(full_dataset):.1%}",
        "100.0%"
    ]
})
print('\n' + '='*50)
print(' SPLIT DE DATOS (80-20)')
print('='*50)
print(split_info.to_string(index=False))
print('='*50)

# Crear DataLoaders
train_loader = DataLoader(
    train_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=True,
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

```

```

val_loader = DataLoader(
    val_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=False, # No shuffle para validación
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

# Resumen de DataLoaders
loader_info = pd.DataFrame({
    'DataLoader': ['Train', 'Validation'],
    'Batches': [len(train_loader), len(val_loader)],
    'Batch Size': [CONFIG['batch_size'], CONFIG['batch_size']],
    'Shuffle': ['Sí', 'No']
})

print('\n' + '='*50)
print(' DATALOADERS')
print('='*50)
print(loader_info.to_string(index=False))
print('='*50)

```

```

=====
DATASET DE ACTIVACIONES
=====

```

	Propiedad	Valor
	Total muestras	59,000
	Shape (59000, 512)	
	Dtype	float32
Tamaño en memoria		115.23 MB

```

=====

=====
SPLIT DE DATOS (80-20)
=====

```

	Conjunto	Muestras	Porcentaje
	Train	47,200	80.0%
Validation		11,800	20.0%
	Total	59,000	100.0%

```

=====

```


=====				
DATALOADERS				
=====				
Loader	Batches	Batch Size	Shuffle	
Train	185	256	Sí	
Validation	47	256	No	
=====				

5. Modelo: Sparse Autoencoder

Modelo importado desde `sae.models.sae.SparseAutoencoder`

Arquitectura:

- **Encoder:** Linear $\rightarrow$ ReLU (sparsity natural)- **Decoder:** Linear (reconstrucción)

```
# Crear modelo desde el módulo
model = SparseAutoencoder(
    input_dim=CONFIG['input_dim'],
    hidden_dim=CONFIG['hidden_dim'],
    tied_weights=CONFIG['tied_weights']
).to(device)

# Calcular parámetros
num_params = sum(p.numel() for p in model.parameters())
encoder_params = sum(p.numel() for p in model.encoder.parameters())
if CONFIG['tied_weights']:
    decoder_params = CONFIG['input_dim'] # Solo bias
else:
    decoder_params = sum(p.numel() for p in model.decoder.parameters())

# Resumen del modelo con formato profesional
model_info = pd.DataFrame({
    'Componente': ['Input', 'Encoder', 'Hidden (Latent)', 'Decoder', 'Output', 'Total'],
    'Dimensión': [
        CONFIG['input_dim'],
        f"{CONFIG['input_dim']} → {CONFIG['hidden_dim']}",
        CONFIG['hidden_dim'],
        f"{CONFIG['hidden_dim']} → {CONFIG['input_dim']}",
        CONFIG['input_dim'],
        '_'
    ],
    'Parámetros': [

```

```

        '-',
        f"{encoder_params:,}",
        '-',
        f"{decoder_params:,}",
        '-',
        f"{num_params:,}"
    ]
})

print('\n' + '='*70)
print(' ARQUITECTURA DEL MODELO')
print('='*70)
print(model_info.to_string(index=False))
print('='*70)
print(f'\n Factor de expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f' Tied weights: {"Sí" if CONFIG["tied_weights"] else "No"}')
print(f' Device: {device}')
print('='*70)

```

```

=====
ARQUITECTURA DEL MODELO
=====

```

	Componente	Dimensión	Parámetros
	Input	512	-
	Encoder	512 → 16384	8,404,992
Hidden	(Latent)	16384	-
	Decoder	16384 → 512	8,389,120
	Output	512	-
	Total	-	16,794,112

```

=====

Factor de expansión: 32.0x
Tied weights: No
Device: cuda
=====

```

6. Función de Pérdida

Loss = MSE(reconstrucción) + λ × L1(activaciones)

- **MSE:** Mide qué tan bien reconstruimos las activaciones originales
- **L1:** Penaliza activaciones grandes → fuerza sparsity

```
def loss_function(x, reconstruction, hidden, sparsity_coef):
    """
    Pérdida compuesta: MSE + L1 penalty

    Args:
        x: Activaciones originales
        reconstruction: Activaciones reconstruidas
        hidden: Activaciones latentes (sparse)
        sparsity_coef: Peso de la penalización L1
    """
    # Error de reconstrucción
    mse_loss = F.mse_loss(reconstruction, x)

    # Sumar sobre las features (dim=1), promediar sobre el batch
    sparsity_loss = hidden.abs().sum(dim=1).mean()

    # Pérdida total
    total_loss = mse_loss + sparsity_coef * sparsity_loss

    return total_loss, mse_loss, sparsity_loss

# Optimizador
optimizer = torch.optim.Adam(model.parameters(), lr=CONFIG['learning_rate'])

# Resumen del optimizador
optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning Rate', 'Sparsity Coef (inicial)', 'Sparsity Decay'],
    'Valor': [
        'Adam',
        f"{CONFIG['learning_rate']:.0e}",
        f"{CONFIG['sparsity_coef']:.0e}",
        f"{CONFIG['sparsity_decay']:.2f}"
    ]
})

print('\n' + '='*50)
print(' OPTIMIZACIÓN')
print('='*50)
print(optim_info.to_string(index=False))
print('='*50)
```

```
=====
OPTIMIZACIÓN
=====
                Parámetro Valor
            Optimizador  Adam
        Learning Rate 1e-03
Sparsity Coef (inicial) 1e-02
        Sparsity Decay  0.98
=====
```

7. Loop de Entrenamiento

```
# Historial de entrenamiento
history = {
    'train_total_loss': [],
    'train_mse_loss': [],
    'train_sparsity_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_total_loss': [],
    'val_mse_loss': [],
    'val_sparsity_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
    'sparsity_coef_history': [] # Guardar evolución del coeficiente L1
}

# Early stopping variables
best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'\n Iniciando entrenamiento: {CONFIG["num_epochs"]} épocas máximo')
print(f' L1 decay: {CONFIG["sparsity_coef"]:.2e} × {CONFIG["sparsity_decay"]:.2f}^época')
print(f' Early stopping: patience={CONFIG["patience"]} épocas')
print('=' * 70)

for epoch in range(CONFIG['num_epochs']):
```

```

# ===== L1 DECAY =====
# Reducir sparsity_coef exponencialmente
current_sparsity_coef = CONFIG['sparsity_coef'] * (CONFIG['sparsity_decay'] ** epoch)
history['sparsity_coef_history'].append(current_sparsity_coef)

# ===== ENTRENAMIENTO =====
model.train()

epoch_total_loss = 0
epoch_mse_loss = 0
epoch_sparsity_loss = 0
epoch_l0 = 0
epoch_fraction_active = 0

# Barra de progreso
pbar = tqdm(train_loader, desc=f'Época {epoch+1}/{CONFIG["num_epochs"]} [Train] L1={current_sparsity_coef}')

for batch in pbar:
    # Mover al device
    x = batch.to(device)

    # Forward pass
    reconstruction, hidden = model(x)

    # Calcular pérdida con sparsity_coef actualizado
    total_loss, mse_loss, sparsity_loss = loss_function(
        x, reconstruction, hidden, current_sparsity_coef
    )

    # Backward pass
    optimizer.zero_grad()
    total_loss.backward()

    optimizer.step()

    # Métricas
    with torch.no_grad():
        l0 = (hidden > 0).float().sum(dim=1).mean().item()
        fraction_active = (hidden > 0).float().mean().item()

    # Acumular
    epoch_total_loss += total_loss.item()

```

```

epoch_mse_loss += mse_loss.item()
epoch_sparsity_loss += sparsity_loss.item()
epoch_l0 += l0
epoch_fraction_active += fraction_active

# Actualizar barra
pbar.set_postfix({
    'loss': f'{total_loss.item():.4f}',
    'mse': f'{mse_loss.item():.4f}',
    'l0': f'{l0:.1f}'
})

# Promedios de entrenamiento
num_train_batches = len(train_loader)
avg_train_total_loss = epoch_total_loss / num_train_batches
avg_train_mse_loss = epoch_mse_loss / num_train_batches
avg_train_sparsity_loss = epoch_sparsity_loss / num_train_batches
avg_train_l0 = epoch_l0 / num_train_batches
avg_train_fraction_active = epoch_fraction_active / num_train_batches

# ===== VALIDACIÓN =====
model.eval()

val_total_loss = 0
val_mse_loss = 0
val_sparsity_loss = 0
val_l0 = 0
val_fraction_active = 0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)

        # Forward pass
        reconstruction, hidden = model(x)

        # Calcular pérdida con sparsity_coef actualizado
        total_loss, mse_loss, sparsity_loss = loss_function(
            x, reconstruction, hidden, current_sparsity_coef
        )

# Métricas

```

```

10 = (hidden > 0).float().sum(dim=1).mean().item()
fraction_active = (hidden > 0).float().mean().item()

# Acumular
val_total_loss += total_loss.item()
val_mse_loss += mse_loss.item()
val_sparsity_loss += sparsity_loss.item()
val_l0 += l0
val_fraction_active += fraction_active

# Promedios de validación
num_val_batches = len(val_loader)
avg_val_total_loss = val_total_loss / num_val_batches
avg_val_mse_loss = val_mse_loss / num_val_batches
avg_val_sparsity_loss = val_sparsity_loss / num_val_batches
avg_val_l0 = val_l0 / num_val_batches
avg_val_fraction_active = val_fraction_active / num_val_batches

# Guardar en historial
history['train_total_loss'].append(avg_train_total_loss)
history['train_mse_loss'].append(avg_train_mse_loss)
history['train_sparsity_loss'].append(avg_train_sparsity_loss)
history['train_l0_sparsity'].append(avg_train_l0)
history['train_fraction_active'].append(avg_train_fraction_active)

history['val_total_loss'].append(avg_val_total_loss)
history['val_mse_loss'].append(avg_val_mse_loss)
history['val_sparsity_loss'].append(avg_val_sparsity_loss)
history['val_l0_sparsity'].append(avg_val_l0)
history['val_fraction_active'].append(avg_val_fraction_active)

# ===== EARLY STOPPING =====
if CONFIG['early_stopping']:
    if avg_val_mse_loss < best_val_mse:
        best_val_mse = avg_val_mse_loss
        best_epoch = epoch + 1
        epochs_no_improve = 0

    # Guardar mejor modelo (local + organizado)
    best_checkpoint = {
        'model_state_dict': model.state_dict(),
        'optimizer_state_dict': optimizer.state_dict(),

```

```

        'config': CONFIG,
        'history': history,
        'epoch': best_epoch,
        'val_mse': best_val_mse
    }

    # Guardado local
    best_model_name = get_model_name(
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best'
    )
    torch.save(best_checkpoint, CONFIG['save_dir'] / best_model_name)

    # Guardado organizado
    save_checkpoint(
        model,
        base_path=NAMING_META['base_path'],
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=CONFIG,
        history=history,
        optimizer=optimizer
    )

    print(f' Mejor modelo guardado en época {best_epoch} (Val MSE: {best_val_mse:.4f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= CONFIG['patience']:
        print(f'\n Early stopping activado en época {epoch+1}')
        print(f' Mejor modelo: época {best_epoch} con Val MSE = {best_val_mse:.4f}')
        break

# Log cada 5 épocas
if (epoch + 1) % 5 == 0 or epoch == 0:
    print(f'\n Época {epoch+1}/{CONFIG["num_epochs"]} | L1 coef: {current_sparsity_coef:}')

```



```

        print(f'  Train - Total: {avg_train_total_loss:.4f} | MSE: {avg_train_mse_loss:.4f}')
        print(f'  Val   - Total: {avg_val_total_loss:.4f} | MSE: {avg_val_mse_loss:.4f} | L0')
        if CONFIG['early_stopping']:
            print(f'    Best Val MSE: {best_val_mse:.4f} @ época {best_epoch} | Sin mejora: {best_val_mse:.4f}')

print('\n' + '='*70)
print(' RESUMEN DEL ENTRENAMIENTO')
print('='*70)

if CONFIG['early_stopping']:
    print(f'Mejor modelo en época {best_epoch} con Val MSE = {best_val_mse:.4f}')
    print(f'Early stopping tras {len(history["train_total_loss"])} épocas')
else:
    print(f'Entrenamiento completo: {CONFIG["num_epochs"]} épocas')

# Tabla de métricas finales
final_metrics = pd.DataFrame({
    'Métrica': ['Total Loss', 'MSE Loss', 'L0 Sparsity', 'Fraction Active'],
    'Train': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['train_fraction_active'][-1]:.2%}"
    ],
    'Validation': [
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['val_l0_sparsity'][-1]:.1f}",
        f"{history['val_fraction_active'][-1]:.2%}"
    ]
})

print('\nMétricas Finales (Última Época):')
print(final_metrics.to_string(index=False))
print('='*70)

```

Iniciando entrenamiento: 250 épocas máximo
 L1 decay: $1.00e-02 \times 0.98^{\text{época}}$
 Early stopping: patience=20 épocas

=====

Época 1/250 [Train] L1=1.00e-02: 100%| | 185/185 [00:02<00:00, 81.38it/s, loss=0.6927, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 1 (Val MSE: 0.6196)

Época 1/250 | L1 coef: 1.00e-02

Train - Total: 0.9509 | MSE: 0.7203 | L0: 152.6

Val - Total: 0.6623 | MSE: 0.6196 | L0: 4.4

Best Val MSE: 0.6196 @ época 1 | Sin mejora: 0/20

Época 2/250 [Train] L1=9.80e-03: 100%| | 185/185 [00:01<00:00, 93.25it/s, loss=0.6309, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 2 (Val MSE: 0.5791)

Época 3/250 [Train] L1=9.60e-03: 100%| | 185/185 [00:01<00:00, 92.76it/s, loss=0.5970, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 3 (Val MSE: 0.5627)

Época 4/250 [Train] L1=9.41e-03: 100%| | 185/185 [00:01<00:00, 93.59it/s, loss=0.5897, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 4 (Val MSE: 0.5475)

Época 5/250 [Train] L1=9.22e-03: 100%| | 185/185 [00:01<00:00, 94.61it/s, loss=0.5836, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 5 (Val MSE: 0.5345)

Época 5/250 | L1 coef: 9.22e-03

Train - Total: 0.5765 | MSE: 0.5408 | L0: 6.1

Val - Total: 0.5715 | MSE: 0.5345 | L0: 9.2

Best Val MSE: 0.5345 @ época 5 | Sin mejora: 0/20

Época 6/250 [Train] L1=9.04e-03: 100%| | 185/185 [00:01<00:00, 95.02it/s, loss=0.5534, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 6 (Val MSE: 0.5246)

Época 7/250 [Train] L1=8.86e-03: 100%| | 185/185 [00:01<00:00, 95.77it/s, loss=0.5602, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 7 (Val MSE: 0.5167)

Época 8/250 [Train] L1=8.68e-03: 100%| | 185/185 [00:01<00:00, 96.05it/s, loss=0.5276, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 8 (Val MSE: 0.5107)

Época 9/250 [Train] L1=8.51e-03: 100%| | 185/185 [00:01<00:00, 96.14it/s, loss=0.5356, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 9 (Val MSE: 0.5072)

Época 10/250 [Train] L1=8.34e-03: 100%| | 185/185 [00:01<00:00, 96.12it/s, loss=0.5543,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 10 (Val MSE: 0.5023)

Época 10/250 | L1 coef: 8.34e-03

Train - Total: 0.5393 | MSE: 0.5043 | L0: 13.4

Val - Total: 0.5380 | MSE: 0.5023 | L0: 16.0

Best Val MSE: 0.5023 @ época 10 | Sin mejora: 0/20

Época 11/250 [Train] L1=8.17e-03: 100%| | 185/185 [00:01<00:00, 96.10it/s, loss=0.5215,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 11 (Val MSE: 0.5000)

Época 12/250 [Train] L1=8.01e-03: 100%| | 185/185 [00:01<00:00, 96.45it/s, loss=0.5308,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 12 (Val MSE: 0.4973)

Época 13/250 [Train] L1=7.85e-03: 100%| | 185/185 [00:01<00:00, 93.66it/s, loss=0.5241,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 13 (Val MSE: 0.4956)

Época 14/250 [Train] L1=7.69e-03: 100%| | 185/185 [00:01<00:00, 92.73it/s, loss=0.5287,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 14 (Val MSE: 0.4938)

Época 15/250 [Train] L1=7.54e-03: 100%| | 185/185 [00:02<00:00, 88.52it/s, loss=0.5257,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 15 (Val MSE: 0.4922)

Época 15/250 | L1 coef: 7.54e-03

Train - Total: 0.5233 | MSE: 0.4925 | L0: 17.0

Val - Total: 0.5232 | MSE: 0.4922 | L0: 19.3

Best Val MSE: 0.4922 @ época 15 | Sin mejora: 0/20

Época 16/250 [Train] L1=7.39e-03: 100%| | 185/185 [00:02<00:00, 92.35it/s, loss=0.5117,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 16 (Val MSE: 0.4906)

Época 17/250 [Train] L1=7.24e-03: 100%| | 185/185 [00:02<00:00, 91.59it/s, loss=0.5331,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 17 (Val MSE: 0.4895)

Época 18/250 [Train] L1=7.09e-03: 100%| | 185/185 [00:01<00:00, 93.68it/s, loss=0.5307,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 18 (Val MSE: 0.4884)

Época 19/250 [Train] L1=6.95e-03: 100%| | 185/185 [00:02<00:00, 91.77it/s, loss=0.5062,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 19 (Val MSE: 0.4872)

Época 20/250 [Train] L1=6.81e-03: 100%| | 185/185 [00:02<00:00, 91.11it/s, loss=0.5164,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 20 (Val MSE: 0.4852)

Época 20/250 | L1 coef: 6.81e-03

Train - Total: 0.5131 | MSE: 0.4860 | L0: 19.3

Val - Total: 0.5127 | MSE: 0.4852 | L0: 21.6

Best Val MSE: 0.4852 @ época 20 | Sin mejora: 0/20

Época 21/250 [Train] L1=6.68e-03: 100%| | 185/185 [00:02<00:00, 81.84it/s, loss=0.4997,

Época 22/250 [Train] L1=6.54e-03: 100%| | 185/185 [00:02<00:00, 92.02it/s, loss=0.4999,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 22 (Val MSE: 0.4834)

Época 23/250 [Train] L1=6.41e-03: 100%| | 185/185 [00:02<00:00, 77.61it/s, loss=0.5142,

Época 24/250 [Train] L1=6.28e-03: 100%| | 185/185 [00:02<00:00, 78.11it/s, loss=0.5032,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 24 (Val MSE: 0.4825)

Época 25/250 [Train] L1=6.16e-03: 100%| | 185/185 [00:02<00:00, 72.73it/s, loss=0.4989,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 25 (Val MSE: 0.4814)

Época 25/250 | L1 coef: 6.16e-03
Train - Total: 0.5062 | MSE: 0.4823 | L0: 20.8
Val - Total: 0.5057 | MSE: 0.4814 | L0: 23.1
Best Val MSE: 0.4814 @ época 25 | Sin mejora: 0/20

Época 26/250 [Train] L1=6.03e-03: 100%| | 185/185 [00:02<00:00, 76.70it/s, loss=0.5124,
Época 27/250 [Train] L1=5.91e-03: 100%| | 185/185 [00:02<00:00, 75.28it/s, loss=0.4977,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 27 (Val MSE: 0.4808)

Época 28/250 [Train] L1=5.80e-03: 100%| | 185/185 [00:02<00:00, 79.07it/s, loss=0.5009,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 28 (Val MSE: 0.4801)

Época 29/250 [Train] L1=5.68e-03: 100%| | 185/185 [00:01<00:00, 94.17it/s, loss=0.4859,
Época 30/250 [Train] L1=5.57e-03: 100%| | 185/185 [00:01<00:00, 92.67it/s, loss=0.5186,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 30 (Val MSE: 0.4789)

Época 30/250 | L1 coef: 5.57e-03
Train - Total: 0.5013 | MSE: 0.4802 | L0: 21.8
Val - Total: 0.5003 | MSE: 0.4789 | L0: 24.1
Best Val MSE: 0.4789 @ época 30 | Sin mejora: 0/20

Época 31/250 [Train] L1=5.45e-03: 100%| | 185/185 [00:02<00:00, 92.42it/s, loss=0.5055,
Época 32/250 [Train] L1=5.35e-03: 100%| | 185/185 [00:02<00:00, 91.56it/s, loss=0.4886,
Época 33/250 [Train] L1=5.24e-03: 100%| | 185/185 [00:01<00:00, 92.89it/s, loss=0.5091,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 33 (Val MSE: 0.4775)

Época 34/250 [Train] L1=5.13e-03: 100%| | 185/185 [00:01<00:00, 93.97it/s, loss=0.4921,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 34 (Val MSE: 0.4771)

Época 35/250 [Train] L1=5.03e-03: 100%| | 185/185 [00:01<00:00, 92.95it/s, loss=0.4842,

Época 35/250 | L1 coef: 5.03e-03

Train - Total: 0.4974 | MSE: 0.4786 | L0: 22.7

Val - Total: 0.4978 | MSE: 0.4787 | L0: 24.9

Best Val MSE: 0.4771 @ época 34 | Sin mejora: 1/20

Época 36/250 [Train] L1=4.93e-03: 100%| | 185/185 [00:01<00:00, 93.67it/s, loss=0.4915,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 36 (Val MSE: 0.4770)

Época 37/250 [Train] L1=4.83e-03: 100%| | 185/185 [00:01<00:00, 92.98it/s, loss=0.4990,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 37 (Val MSE: 0.4767)

Época 38/250 [Train] L1=4.74e-03: 100%| | 185/185 [00:01<00:00, 92.84it/s, loss=0.5074,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 38 (Val MSE: 0.4761)

Época 39/250 [Train] L1=4.64e-03: 100%| | 185/185 [00:01<00:00, 94.07it/s, loss=0.5128,

Época 40/250 [Train] L1=4.55e-03: 100%| | 185/185 [00:02<00:00, 92.32it/s, loss=0.4936,

Época 40/250 | L1 coef: 4.55e-03

Train - Total: 0.4936 | MSE: 0.4767 | L0: 23.5

Val - Total: 0.4936 | MSE: 0.4764 | L0: 25.6

Best Val MSE: 0.4761 @ época 38 | Sin mejora: 2/20

Época 41/250 [Train] L1=4.46e-03: 100%| | 185/185 [00:02<00:00, 91.83it/s, loss=0.5057,
Época 42/250 [Train] L1=4.37e-03: 100%| | 185/185 [00:02<00:00, 92.25it/s, loss=0.5018,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 42 (Val MSE: 0.4755)

Época 43/250 [Train] L1=4.28e-03: 100%| | 185/185 [00:01<00:00, 92.82it/s, loss=0.5007,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 43 (Val MSE: 0.4751)

Época 44/250 [Train] L1=4.19e-03: 100%| | 185/185 [00:01<00:00, 93.86it/s, loss=0.4959,
Época 45/250 [Train] L1=4.11e-03: 100%| | 185/185 [00:02<00:00, 92.39it/s, loss=0.5011,

Época 45/250 | L1 coef: 4.11e-03

Train - Total: 0.4914 | MSE: 0.4762 | L0: 24.0

Val - Total: 0.4910 | MSE: 0.4756 | L0: 26.2

Best Val MSE: 0.4751 @ época 43 | Sin mejora: 2/20

Época 46/250 [Train] L1=4.03e-03: 100%| | 185/185 [00:01<00:00, 92.76it/s, loss=0.4811,
Época 47/250 [Train] L1=3.95e-03: 100%| | 185/185 [00:01<00:00, 92.59it/s, loss=0.4916,
Época 48/250 [Train] L1=3.87e-03: 100%| | 185/185 [00:01<00:00, 94.15it/s, loss=0.4900,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 48 (Val MSE: 0.4749)

Época 49/250 [Train] L1=3.79e-03: 100%| | 185/185 [00:01<00:00, 93.27it/s, loss=0.5021,
Época 50/250 [Train] L1=3.72e-03: 100%| | 185/185 [00:01<00:00, 94.02it/s, loss=0.4686,

Época 50/250 | L1 coef: 3.72e-03

Train - Total: 0.4903 | MSE: 0.4767 | L0: 24.3

Val - Total: 0.4892 | MSE: 0.4753 | L0: 26.6

Best Val MSE: 0.4749 @ época 48 | Sin mejora: 2/20

Época 51/250 [Train] L1=3.64e-03: 100%| | 185/185 [00:01<00:00, 93.15it/s, loss=0.4969,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 51 (Val MSE: 0.4739)

Época 52/250 [Train] L1=3.57e-03: 100%| | 185/185 [00:01<00:00, 92.56it/s, loss=0.4694,
Época 53/250 [Train] L1=3.50e-03: 100%| | 185/185 [00:01<00:00, 93.03it/s, loss=0.4876,
Época 54/250 [Train] L1=3.43e-03: 100%| | 185/185 [00:01<00:00, 93.22it/s, loss=0.5050,
Época 55/250 [Train] L1=3.36e-03: 100%| | 185/185 [00:01<00:00, 93.32it/s, loss=0.4737,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 55 (Val MSE: 0.4731)

Época 55/250 | L1 coef: 3.36e-03
Train - Total: 0.4878 | MSE: 0.4753 | L0: 24.8
Val - Total: 0.4859 | MSE: 0.4731 | L0: 27.1
Best Val MSE: 0.4731 @ época 55 | Sin mejora: 0/20

Época 56/250 [Train] L1=3.29e-03: 100%| | 185/185 [00:01<00:00, 92.62it/s, loss=0.4975,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 56 (Val MSE: 0.4728)

Época 57/250 [Train] L1=3.23e-03: 100%| | 185/185 [00:01<00:00, 94.42it/s, loss=0.4864,
Época 58/250 [Train] L1=3.16e-03: 100%| | 185/185 [00:01<00:00, 94.05it/s, loss=0.4874,
Época 59/250 [Train] L1=3.10e-03: 100%| | 185/185 [00:01<00:00, 94.34it/s, loss=0.4796,
Época 60/250 [Train] L1=3.04e-03: 100%| | 185/185 [00:01<00:00, 93.74it/s, loss=0.4978,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 60 (Val MSE: 0.4723)

Época 60/250 | L1 coef: 3.04e-03
Train - Total: 0.4856 | MSE: 0.4743 | L0: 25.1
Val - Total: 0.4838 | MSE: 0.4723 | L0: 27.4
Best Val MSE: 0.4723 @ época 60 | Sin mejora: 0/20

Época 61/250 [Train] L1=2.98e-03: 100%| | 185/185 [00:02<00:00, 91.69it/s, loss=0.4729,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 61 (Val MSE: 0.4722)

Época 62/250 [Train]	L1=2.92e-03: 100%	185/185 [00:01<00:00, 93.06it/s, loss=0.4818,
Época 63/250 [Train]	L1=2.86e-03: 100%	185/185 [00:01<00:00, 93.27it/s, loss=0.4895,
Época 64/250 [Train]	L1=2.80e-03: 100%	185/185 [00:01<00:00, 94.32it/s, loss=0.4835,
Época 65/250 [Train]	L1=2.74e-03: 100%	185/185 [00:01<00:00, 92.70it/s, loss=0.4700,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 65 (Val MSE: 0.4714)

Época 65/250 | L1 coef: 2.74e-03

Train - Total: 0.4833 | MSE: 0.4730 | L0: 25.6

Val - Total: 0.4820 | MSE: 0.4714 | L0: 27.9

Best Val MSE: 0.4714 @ época 65 | Sin mejora: 0/20

Época 66/250 [Train]	L1=2.69e-03: 100%	185/185 [00:02<00:00, 90.12it/s, loss=0.5027,
Época 67/250 [Train]	L1=2.64e-03: 100%	185/185 [00:01<00:00, 93.01it/s, loss=0.4661,
Época 68/250 [Train]	L1=2.58e-03: 100%	185/185 [00:02<00:00, 92.29it/s, loss=0.5025,
Época 69/250 [Train]	L1=2.53e-03: 100%	185/185 [00:01<00:00, 92.75it/s, loss=0.4714,
Época 70/250 [Train]	L1=2.48e-03: 100%	185/185 [00:01<00:00, 92.76it/s, loss=0.4907,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 70 (Val MSE: 0.4711)

Época 70/250 | L1 coef: 2.48e-03

Train - Total: 0.4818 | MSE: 0.4724 | L0: 25.9

Val - Total: 0.4808 | MSE: 0.4711 | L0: 28.1

Best Val MSE: 0.4711 @ época 70 | Sin mejora: 0/20

Época 71/250 [Train]	L1=2.43e-03: 100%	185/185 [00:02<00:00, 91.85it/s, loss=0.4651,
Época 72/250 [Train]	L1=2.38e-03: 100%	185/185 [00:01<00:00, 94.03it/s, loss=0.4729,
Época 73/250 [Train]	L1=2.33e-03: 100%	185/185 [00:01<00:00, 94.04it/s, loss=0.4626,
Época 74/250 [Train]	L1=2.29e-03: 100%	185/185 [00:01<00:00, 94.04it/s, loss=0.4850,
Época 75/250 [Train]	L1=2.24e-03: 100%	185/185 [00:02<00:00, 91.89it/s, loss=0.4856,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 75 (Val MSE: 0.4710)

Época 75/250 | L1 coef: 2.24e-03

Train - Total: 0.4813 | MSE: 0.4727 | L0: 26.0

Val - Total: 0.4798 | MSE: 0.4710 | L0: 28.3

Best Val MSE: 0.4710 @ época 75 | Sin mejora: 0/20

Época 76/250 [Train] L1=2.20e-03: 100%| | 185/185 [00:02<00:00, 91.85it/s, loss=0.4874,

Época 77/250 [Train] L1=2.15e-03: 100%| | 185/185 [00:01<00:00, 92.67it/s, loss=0.4693,

Época 78/250 [Train] L1=2.11e-03: 100%| | 185/185 [00:02<00:00, 92.33it/s, loss=0.4883,

Época 79/250 [Train] L1=2.07e-03: 100%| | 185/185 [00:01<00:00, 93.05it/s, loss=0.4956,

Época 80/250 [Train] L1=2.03e-03: 100%| | 185/185 [00:02<00:00, 92.26it/s, loss=0.4838,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 80 (Val MSE: 0.4703)

Época 80/250 | L1 coef: 2.03e-03

Train - Total: 0.4800 | MSE: 0.4721 | L0: 26.3

Val - Total: 0.4784 | MSE: 0.4703 | L0: 28.5

Best Val MSE: 0.4703 @ época 80 | Sin mejora: 0/20

Época 81/250 [Train] L1=1.99e-03: 100%| | 185/185 [00:02<00:00, 90.95it/s, loss=0.4964,

Época 82/250 [Train] L1=1.95e-03: 100%| | 185/185 [00:01<00:00, 92.67it/s, loss=0.4958,

Época 83/250 [Train] L1=1.91e-03: 100%| | 185/185 [00:01<00:00, 93.12it/s, loss=0.4710,

Época 84/250 [Train] L1=1.87e-03: 100%| | 185/185 [00:01<00:00, 92.87it/s, loss=0.4999,

Época 85/250 [Train] L1=1.83e-03: 100%| | 185/185 [00:02<00:00, 92.40it/s, loss=0.4886,

Época 85/250 | L1 coef: 1.83e-03

Train - Total: 0.4798 | MSE: 0.4726 | L0: 26.4

Val - Total: 0.4801 | MSE: 0.4728 | L0: 28.5

Best Val MSE: 0.4703 @ época 80 | Sin mejora: 5/20

Época 86/250 [Train] L1=1.80e-03: 100%| | 185/185 [00:01<00:00, 93.84it/s, loss=0.5038,

Época 87/250 [Train] L1=1.76e-03: 100%| | 185/185 [00:02<00:00, 92.19it/s, loss=0.4988,

Época 88/250 [Train] L1=1.72e-03: 100%| | 185/185 [00:01<00:00, 93.20it/s, loss=0.4705,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 88 (Val MSE: 0.4697)

Época 89/250 [Train] L1=1.69e-03: 100%| | 185/185 [00:01<00:00, 92.97it/s, loss=0.4883,
Época 90/250 [Train] L1=1.66e-03: 100%| | 185/185 [00:01<00:00, 93.87it/s, loss=0.4859,

Época 90/250 | L1 coef: 1.66e-03

Train - Total: 0.4815 | MSE: 0.4749 | L0: 26.4

Val - Total: 0.4782 | MSE: 0.4714 | L0: 28.8

Best Val MSE: 0.4697 @ época 88 | Sin mejora: 2/20

Época 91/250 [Train] L1=1.62e-03: 100%| | 185/185 [00:01<00:00, 94.09it/s, loss=0.4892,
Época 92/250 [Train] L1=1.59e-03: 100%| | 185/185 [00:01<00:00, 94.26it/s, loss=0.4626,
Época 93/250 [Train] L1=1.56e-03: 100%| | 185/185 [00:01<00:00, 94.51it/s, loss=0.4829,
Época 94/250 [Train] L1=1.53e-03: 100%| | 185/185 [00:01<00:00, 93.28it/s, loss=0.4878,
Época 95/250 [Train] L1=1.50e-03: 100%| | 185/185 [00:01<00:00, 93.48it/s, loss=0.4836,

Época 95/250 | L1 coef: 1.50e-03

Train - Total: 0.4773 | MSE: 0.4711 | L0: 26.8

Val - Total: 0.4784 | MSE: 0.4722 | L0: 28.8

Best Val MSE: 0.4697 @ época 88 | Sin mejora: 7/20

Época 96/250 [Train] L1=1.47e-03: 100%| | 185/185 [00:02<00:00, 92.23it/s, loss=0.4875,
Época 97/250 [Train] L1=1.44e-03: 100%| | 185/185 [00:01<00:00, 93.99it/s, loss=0.4944,
Época 98/250 [Train] L1=1.41e-03: 100%| | 185/185 [00:01<00:00, 93.94it/s, loss=0.4775,
Época 99/250 [Train] L1=1.38e-03: 100%| | 185/185 [00:01<00:00, 93.24it/s, loss=0.4879,
Época 100/250 [Train] L1=1.35e-03: 100%| | 185/185 [00:01<00:00, 94.14it/s, loss=0.4745,

Época 100/250 | L1 coef: 1.35e-03

Train - Total: 0.4769 | MSE: 0.4712 | L0: 26.9

Val - Total: 0.4771 | MSE: 0.4713 | L0: 29.0

Best Val MSE: 0.4697 @ época 88 | Sin mejora: 12/20

Época 101/250 [Train] L1=1.33e-03: 100%| | 185/185 [00:01<00:00, 93.34it/s, loss=0.4908,
Época 102/250 [Train] L1=1.30e-03: 100%| | 185/185 [00:01<00:00, 94.26it/s, loss=0.4678,
Época 103/250 [Train] L1=1.27e-03: 100%| | 185/185 [00:01<00:00, 94.71it/s, loss=0.4578,
Época 104/250 [Train] L1=1.25e-03: 100%| | 185/185 [00:01<00:00, 93.67it/s, loss=0.4751,
Época 105/250 [Train] L1=1.22e-03: 100%| | 185/185 [00:01<00:00, 93.20it/s, loss=0.4647,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 105 (Val MSE: 0.4694)

Época 105/250 | L1 coef: 1.22e-03
Train - Total: 0.4757 | MSE: 0.4706 | L0: 27.0
Val - Total: 0.4748 | MSE: 0.4694 | L0: 29.3
Best Val MSE: 0.4694 @ época 105 | Sin mejora: 0/20

Época 106/250 [Train] L1=1.20e-03: 100%| | 185/185 [00:01<00:00, 92.64it/s, loss=0.4731,
Época 107/250 [Train] L1=1.17e-03: 100%| | 185/185 [00:01<00:00, 93.73it/s, loss=0.4987,
Época 108/250 [Train] L1=1.15e-03: 100%| | 185/185 [00:01<00:00, 93.19it/s, loss=0.4752,
Época 109/250 [Train] L1=1.13e-03: 100%| | 185/185 [00:01<00:00, 93.64it/s, loss=0.4795,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 109 (Val MSE: 0.4694)

Época 110/250 [Train] L1=1.11e-03: 100%| | 185/185 [00:02<00:00, 92.31it/s, loss=0.4698,

Época 110/250 | L1 coef: 1.11e-03
Train - Total: 0.4757 | MSE: 0.4709 | L0: 27.1
Val - Total: 0.4749 | MSE: 0.4700 | L0: 29.3
Best Val MSE: 0.4694 @ época 109 | Sin mejora: 1/20

Época 111/250 [Train] L1=1.08e-03: 100%| | 185/185 [00:01<00:00, 93.09it/s, loss=0.4538,
Época 112/250 [Train] L1=1.06e-03: 100%| | 185/185 [00:01<00:00, 93.50it/s, loss=0.4552,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 112 (Val MSE: 0.4693)

Época 113/250 [Train] L1=1.04e-03: 100%| | 185/185 [00:02<00:00, 92.49it/s, loss=0.4644,
Época 114/250 [Train] L1=1.02e-03: 100%| | 185/185 [00:02<00:00, 92.48it/s, loss=0.4666,
Época 115/250 [Train] L1=9.99e-04: 100%| | 185/185 [00:01<00:00, 93.57it/s, loss=0.4879,

Época 115/250 | L1 coef: 9.99e-04
Train - Total: 0.4760 | MSE: 0.4716 | L0: 27.2
Val - Total: 0.4750 | MSE: 0.4706 | L0: 29.4
Best Val MSE: 0.4693 @ época 112 | Sin mejora: 3/20

Época 116/250 [Train] L1=9.79e-04: 100%| | 185/185 [00:02<00:00, 91.63it/s, loss=0.4892,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 116 (Val MSE: 0.4691)

Época 117/250 [Train] L1=9.60e-04: 100%| | 185/185 [00:02<00:00, 91.65it/s, loss=0.4673,

Época 118/250 [Train] L1=9.41e-04: 100%| | 185/185 [00:02<00:00, 91.74it/s, loss=0.4868,

Época 119/250 [Train] L1=9.22e-04: 100%| | 185/185 [00:01<00:00, 92.69it/s, loss=0.4957,

Época 120/250 [Train] L1=9.03e-04: 100%| | 185/185 [00:01<00:00, 92.91it/s, loss=0.4703,

Época 120/250 | L1 coef: 9.03e-04

Train - Total: 0.4755 | MSE: 0.4714 | L0: 27.3

Val - Total: 0.4765 | MSE: 0.4724 | L0: 29.4

Best Val MSE: 0.4691 @ época 116 | Sin mejora: 4/20

Época 121/250 [Train] L1=8.85e-04: 100%| | 185/185 [00:01<00:00, 92.94it/s, loss=0.4477,

Época 122/250 [Train] L1=8.68e-04: 100%| | 185/185 [00:01<00:00, 92.50it/s, loss=0.4887,

Época 123/250 [Train] L1=8.50e-04: 100%| | 185/185 [00:01<00:00, 93.44it/s, loss=0.4853,

Época 124/250 [Train] L1=8.33e-04: 100%| | 185/185 [00:01<00:00, 93.41it/s, loss=0.4825,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt

Mejor modelo guardado en época 124 (Val MSE: 0.4691)

Época 125/250 [Train] L1=8.17e-04: 100%| | 185/185 [00:02<00:00, 92.27it/s, loss=0.4907,

Época 125/250 | L1 coef: 8.17e-04

Train - Total: 0.4733 | MSE: 0.4695 | L0: 27.5

Val - Total: 0.4734 | MSE: 0.4696 | L0: 29.7

Best Val MSE: 0.4691 @ época 124 | Sin mejora: 1/20

Época 126/250 [Train] L1=8.00e-04: 100%| | 185/185 [00:02<00:00, 92.33it/s, loss=0.4831,

Época 127/250 [Train] L1=7.84e-04: 100%| | 185/185 [00:02<00:00, 92.34it/s, loss=0.4724,

Época 128/250 [Train] L1=7.69e-04: 100%| | 185/185 [00:01<00:00, 93.76it/s, loss=0.4830,

Época 129/250 [Train] L1=7.53e-04: 100%| | 185/185 [00:01<00:00, 92.51it/s, loss=0.4896,

Época 130/250 [Train] L1=7.38e-04: 100%| | 185/185 [00:01<00:00, 92.95it/s, loss=0.4746,

Época 130/250 | L1 coef: 7.38e-04
Train - Total: 0.4741 | MSE: 0.4706 | L0: 27.5
Val - Total: 0.4727 | MSE: 0.4692 | L0: 29.7
Best Val MSE: 0.4691 @ época 124 | Sin mejora: 6/20

Época 131/250 [Train] L1=7.23e-04: 100%| | 185/185 [00:01<00:00, 92.63it/s, loss=0.4869,
Época 132/250 [Train] L1=7.09e-04: 100%| | 185/185 [00:01<00:00, 94.18it/s, loss=0.4833,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 132 (Val MSE: 0.4688)

Época 133/250 [Train] L1=6.95e-04: 100%| | 185/185 [00:01<00:00, 92.55it/s, loss=0.4645,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
Mejor modelo guardado en época 133 (Val MSE: 0.4685)

Época 134/250 [Train] L1=6.81e-04: 100%| | 185/185 [00:02<00:00, 91.70it/s, loss=0.4796,
Época 135/250 [Train] L1=6.67e-04: 100%| | 185/185 [00:02<00:00, 91.28it/s, loss=0.5340,

Época 135/250 | L1 coef: 6.67e-04
Train - Total: 0.4762 | MSE: 0.4731 | L0: 27.5
Val - Total: 0.4930 | MSE: 0.4899 | L0: 29.3
Best Val MSE: 0.4685 @ época 133 | Sin mejora: 2/20

Época 136/250 [Train] L1=6.54e-04: 100%| | 185/185 [00:02<00:00, 92.48it/s, loss=0.4756,
Época 137/250 [Train] L1=6.41e-04: 100%| | 185/185 [00:01<00:00, 93.06it/s, loss=0.4647,
Época 138/250 [Train] L1=6.28e-04: 100%| | 185/185 [00:02<00:00, 91.49it/s, loss=0.4722,
Época 139/250 [Train] L1=6.15e-04: 100%| | 185/185 [00:01<00:00, 93.24it/s, loss=0.4828,
Época 140/250 [Train] L1=6.03e-04: 100%| | 185/185 [00:01<00:00, 93.45it/s, loss=0.4820,

Época 140/250 | L1 coef: 6.03e-04
Train - Total: 0.4731 | MSE: 0.4702 | L0: 27.6
Val - Total: 0.4724 | MSE: 0.4694 | L0: 29.8
Best Val MSE: 0.4685 @ época 133 | Sin mejora: 7/20

Época 141/250 [Train] L1=5.91e-04: 100%| | 185/185 [00:01<00:00, 93.48it/s, loss=0.4641,
 Época 142/250 [Train] L1=5.79e-04: 100%| | 185/185 [00:01<00:00, 92.91it/s, loss=0.4714,
 Época 143/250 [Train] L1=5.68e-04: 100%| | 185/185 [00:02<00:00, 92.42it/s, loss=0.4801,
 Época 144/250 [Train] L1=5.56e-04: 100%| | 185/185 [00:01<00:00, 93.58it/s, loss=0.4651,
 Época 145/250 [Train] L1=5.45e-04: 100%| | 185/185 [00:01<00:00, 93.03it/s, loss=0.4590

Época 145/250 | L1 coef: 5.45e-04
 Train - Total: 0.4725 | MSE: 0.4698 | L0: 27.7
 Val - Total: 0.4727 | MSE: 0.4700 | L0: 29.8
 Best Val MSE: 0.4685 @ época 133 | Sin mejora: 12/20

Época 146/250 [Train] L1=5.34e-04: 100%| | 185/185 [00:01<00:00, 94.12it/s, loss=0.4781,
 Época 147/250 [Train] L1=5.24e-04: 100%| | 185/185 [00:01<00:00, 93.84it/s, loss=0.4646,
 Época 148/250 [Train] L1=5.13e-04: 100%| | 185/185 [00:01<00:00, 93.73it/s, loss=0.4689,
 Época 149/250 [Train] L1=5.03e-04: 100%| | 185/185 [00:01<00:00, 92.59it/s, loss=0.4750,
 Época 150/250 [Train] L1=4.93e-04: 100%| | 185/185 [00:01<00:00, 92.80it/s, loss=0.4829,

Época 150/250 | L1 coef: 4.93e-04
 Train - Total: 0.4719 | MSE: 0.4694 | L0: 27.8
 Val - Total: 0.4716 | MSE: 0.4690 | L0: 29.9
 Best Val MSE: 0.4685 @ época 133 | Sin mejora: 17/20

Época 151/250 [Train] L1=4.83e-04: 100%| | 185/185 [00:01<00:00, 93.71it/s, loss=0.4706,
 Época 152/250 [Train] L1=4.73e-04: 100%| | 185/185 [00:02<00:00, 92.36it/s, loss=0.4426,
 Época 153/250 [Train] L1=4.64e-04: 100%| | 185/185 [00:02<00:00, 91.81it/s, loss=0.4642,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
 Mejor modelo guardado en época 153 (Val MSE: 0.4683)

Época 154/250 [Train] L1=4.55e-04: 100%| | 185/185 [00:02<00:00, 92.19it/s, loss=0.4772,
 Época 155/250 [Train] L1=4.45e-04: 100%| | 185/185 [00:01<00:00, 92.82it/s, loss=0.4738,

Época 155/250 | L1 coef: 4.45e-04
 Train - Total: 0.4744 | MSE: 0.4721 | L0: 27.7
 Val - Total: 0.4731 | MSE: 0.4708 | L0: 29.8
 Best Val MSE: 0.4683 @ época 153 | Sin mejora: 2/20

Época 156/250 [Train] L1=4.37e-04: 100%| | 185/185 [00:01<00:00, 93.63it/s, loss=0.4676,
 Época 157/250 [Train] L1=4.28e-04: 100%| | 185/185 [00:02<00:00, 92.27it/s, loss=0.4584,
 Época 158/250 [Train] L1=4.19e-04: 100%| | 185/185 [00:01<00:00, 93.17it/s, loss=0.4692,
 Época 159/250 [Train] L1=4.11e-04: 100%| | 185/185 [00:02<00:00, 90.62it/s, loss=0.4899,
 Época 160/250 [Train] L1=4.03e-04: 100%| | 185/185 [00:02<00:00, 92.10it/s, loss=0.4804

Época 160/250 | L1 coef: 4.03e-04

Train - Total: 0.4725 | MSE: 0.4704 | L0: 27.8

Val - Total: 0.4707 | MSE: 0.4685 | L0: 30.0

Best Val MSE: 0.4683 @ época 153 | Sin mejora: 7/20

Época 161/250 [Train] L1=3.95e-04: 100%| | 185/185 [00:01<00:00, 93.16it/s, loss=0.4611,
 Época 162/250 [Train] L1=3.87e-04: 100%| | 185/185 [00:01<00:00, 92.90it/s, loss=0.4653,
 Época 163/250 [Train] L1=3.79e-04: 100%| | 185/185 [00:02<00:00, 92.35it/s, loss=0.4488,
 Época 164/250 [Train] L1=3.71e-04: 100%| | 185/185 [00:02<00:00, 91.72it/s, loss=0.4791,
 Época 165/250 [Train] L1=3.64e-04: 100%| | 185/185 [00:01<00:00, 93.15it/s, loss=0.4803,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_best.pt
 Mejor modelo guardado en época 165 (Val MSE: 0.4681)

Época 165/250 | L1 coef: 3.64e-04

Train - Total: 0.4719 | MSE: 0.4700 | L0: 27.9

Val - Total: 0.4701 | MSE: 0.4681 | L0: 30.0

Best Val MSE: 0.4681 @ época 165 | Sin mejora: 0/20

Época 166/250 [Train] L1=3.57e-04: 100%| | 185/185 [00:01<00:00, 92.62it/s, loss=0.4533,
 Época 167/250 [Train] L1=3.50e-04: 100%| | 185/185 [00:01<00:00, 92.81it/s, loss=0.4715,
 Época 168/250 [Train] L1=3.43e-04: 100%| | 185/185 [00:02<00:00, 91.84it/s, loss=0.4644,
 Época 169/250 [Train] L1=3.36e-04: 100%| | 185/185 [00:02<00:00, 92.30it/s, loss=0.4784,
 Época 170/250 [Train] L1=3.29e-04: 100%| | 185/185 [00:02<00:00, 92.34it/s, loss=0.4778,

Época 170/250 | L1 coef: 3.29e-04

Train - Total: 0.4712 | MSE: 0.4694 | L0: 27.9

Val - Total: 0.4706 | MSE: 0.4688 | L0: 30.1

Best Val MSE: 0.4681 @ época 165 | Sin mejora: 5/20

Época 171/250 [Train] L1=3.22e-04: 100%| | 185/185 [00:01<00:00, 93.48it/s, loss=0.4555,
Época 172/250 [Train] L1=3.16e-04: 100%| | 185/185 [00:01<00:00, 92.58it/s, loss=0.4601,
Época 173/250 [Train] L1=3.10e-04: 100%| | 185/185 [00:02<00:00, 92.38it/s, loss=0.4745,
Época 174/250 [Train] L1=3.03e-04: 100%| | 185/185 [00:01<00:00, 92.76it/s, loss=0.4836,
Época 175/250 [Train] L1=2.97e-04: 100%| | 185/185 [00:01<00:00, 92.95it/s, loss=0.4764

Época 175/250 | L1 coef: 2.97e-04

Train - Total: 0.4711 | MSE: 0.4694 | L0: 28.0

Val - Total: 0.4703 | MSE: 0.4686 | L0: 30.1

Best Val MSE: 0.4681 @ época 165 | Sin mejora: 10/20

Época 176/250 [Train] L1=2.91e-04: 100%| | 185/185 [00:01<00:00, 92.71it/s, loss=0.4612,
Época 177/250 [Train] L1=2.86e-04: 100%| | 185/185 [00:01<00:00, 92.92it/s, loss=0.4635,
Época 178/250 [Train] L1=2.80e-04: 100%| | 185/185 [00:01<00:00, 92.83it/s, loss=0.4638,
Época 179/250 [Train] L1=2.74e-04: 100%| | 185/185 [00:01<00:00, 92.52it/s, loss=0.4537,
Época 180/250 [Train] L1=2.69e-04: 100%| | 185/185 [00:02<00:00, 92.46it/s, loss=0.4707,

Época 180/250 | L1 coef: 2.69e-04

Train - Total: 0.4711 | MSE: 0.4696 | L0: 28.0

Val - Total: 0.4705 | MSE: 0.4690 | L0: 30.1

Best Val MSE: 0.4681 @ época 165 | Sin mejora: 15/20

Época 181/250 [Train] L1=2.63e-04: 100%| | 185/185 [00:02<00:00, 92.07it/s, loss=0.4769,
Época 182/250 [Train] L1=2.58e-04: 100%| | 185/185 [00:01<00:00, 93.34it/s, loss=0.4701,
Época 183/250 [Train] L1=2.53e-04: 100%| | 185/185 [00:01<00:00, 93.14it/s, loss=0.4713,
Época 184/250 [Train] L1=2.48e-04: 100%| | 185/185 [00:02<00:00, 92.24it/s, loss=0.4585,
Época 185/250 [Train] L1=2.43e-04: 100%| | 185/185 [00:01<00:00, 93.38it/s, loss=0.4641,

Early stopping activado en época 185

Mejor modelo: época 165 con Val MSE = 0.4681

RESUMEN DEL ENTRENAMIENTO

Mejor modelo en época 165 con Val MSE = 0.4681

Early stopping tras 185 épocas

Métricas Finales (Última Época):

	Métrica	Train	Validation
Total Loss	0.4701		0.4696
MSE Loss	0.4687		0.4682
L0 Sparsity	28.1		30.2
Fraction Active	0.17%		0.18%

=====

8. Guardar Modelo

```
# Guardar checkpoint final (local + organizado)
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': CONFIG,
    'history': history
}

# Guardado local
final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final'
)
checkpoint_path = CONFIG['save_dir'] / final_model_name
torch.save(checkpoint, checkpoint_path)

# Guardado organizado
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=CONFIG,
    history=history,
    optimizer=optimizer
)
```

```

)

# Resumen de archivos guardados
saved_files = pd.DataFrame({
    'Tipo': ['Local', 'Organizado'],
    'Ubicación': [
        str(checkpoint_path),
        f"layer_{NAMING_META['capa']:02d}/models/..."
    ]
})

print('\n' + '='*70)
print(' CHECKPOINTS GUARDADOS')
print('='*70)
print(saved_files.to_string(index=False))
print('='*70)

```

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex5_final.pt

```

=====
CHECKPOINTS GUARDADOS
=====

```

	Tipo	Ubicación
Local	saved_model\sae_standard_l1-decay_l6_1000g_final.pt	
Organizado		layer_06/models/...

```

=====

```

9. Métricas básicas del modelo

```

# Guardar metricas de entrenamiento
metrics_dir = get_metrics_dir(
    NAMING_META["experiment_base_path"],
    NAMING_META["variante"],
    NAMING_META["tecnica"],
    NAMING_META["capa"],
    NAMING_META["juegos"],
    extras_id=extras_id
)

```

```

training_metrics = {
    # Metricas de calidad
    "val_mse_final": history["val_mse_loss"][-1],
    "val_mse_best": best_val_mse,
    "val_l0_final": history["val_l0_sparsity"][-1],
    "val_fraction_active_final": history["val_fraction_active"][-1],
    # Contexto del entrenamiento
    "best_epoch": best_epoch,
    "total_epochs": len(history["train_total_loss"]),
    "train_mse_final": history["train_mse_loss"][-1],
    "train_l0_final": history["train_l0_sparsity"][-1],
    # Hiperparametros
    "hidden_dim": CONFIG["hidden_dim"],
    "expansion_factor": CONFIG["hidden_dim"] / CONFIG["input_dim"],
    "sparsity_coef": CONFIG["sparsity_coef"],
    "sparsity_decay": CONFIG["sparsity_decay"],
    "learning_rate": CONFIG["learning_rate"],
    "num_games": NAMING_META["juegos"],
}

metrics_path = metrics_dir / "training_metrics.json"
with open(metrics_path, "w") as f:
    _json.dump(training_metrics, f, indent=2)

# Resumen
metrics_df = pd.DataFrame({
    "Metrica": ["Val MSE (final)", "Val MSE (mejor)", "Val L0 (final)", "Fraccion Activa"],
    "Valor": [
        f"{training_metrics['val_mse_final']:.6f}",
        f"{training_metrics['val_mse_best']:.6f}",
        f"{training_metrics['val_l0_final']:.1f}",
        f"{training_metrics['val_fraction_active_final']:.2%}"
    ]
})

print("" + "="*50)
print(" METRICAS DE ENTRENAMIENTO GUARDADAS")
print("="*50)
print(metrics_df.to_string(index=False))
print(f"Archivo: {metrics_path}")
print("="*50)

```

=====

METRICAS DE ENTRENAMIENTO GUARDADAS

=====

Metrica	Valor
Val MSE (final)	0.468234
Val MSE (mejor)	0.468089
Val L0 (final)	30.2
Fraccion Activa	0.18%

Archivo: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\decay_standard\1000games\ex5\metrics\training_metrics.json

=====

10. Visualización de Resultados

```
# Gráficas de pérdidas (Train vs Validation) + L1 Decay
fig, axes = plt.subplots(2, 3, figsize=(18, 10))

# Total loss
axes[0, 0].plot(history['train_total_loss'], label='Train', alpha=0.8)
axes[0, 0].plot(history['val_total_loss'], label='Validation', alpha=0.8)
axes[0, 0].set_title('Total Loss')
axes[0, 0].set_xlabel('Época')
axes[0, 0].set_ylabel('Loss')
axes[0, 0].legend()
axes[0, 0].grid(True, alpha=0.3)

# MSE loss
axes[0, 1].plot(history['train_mse_loss'], label='Train', color='orange', alpha=0.8)
axes[0, 1].plot(history['val_mse_loss'], label='Validation', color='red', alpha=0.8)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[0, 1].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best')
axes[0, 1].set_title('MSE Loss (Reconstrucción)')
axes[0, 1].set_xlabel('Época')
axes[0, 1].set_ylabel('MSE')
axes[0, 1].legend()
axes[0, 1].grid(True, alpha=0.3)

# L0 sparsity
axes[0, 2].plot(history['train_l0_sparsity'], label='Train', color='green', alpha=0.8)
axes[0, 2].plot(history['val_l0_sparsity'], label='Validation', color='darkgreen', alpha=0.8)
axes[0, 2].set_title('L0 Sparsity (Features Activas)')
axes[0, 2].set_xlabel('Época')
```

```

axes[0, 2].set_ylabel('Número de features')
axes[0, 2].legend()
axes[0, 2].grid(True, alpha=0.3)

# Fraction active
axes[1, 0].plot(history['train_fraction_active'], label='Train', color='purple', alpha=0.8)
axes[1, 0].plot(history['val_fraction_active'], label='Validation', color='magenta', alpha=0.8)
axes[1, 0].set_title('Fracción de Features Activas')
axes[1, 0].set_xlabel('Época')
axes[1, 0].set_ylabel('Fracción')
axes[1, 0].legend()
axes[1, 0].grid(True, alpha=0.3)

# Sparsity Coefficient (L1 Decay)
axes[1, 1].plot(history['sparsity_coef_history'], color='brown', linewidth=2)
axes[1, 1].set_title('L1 Coefficient (Decay)')
axes[1, 1].set_xlabel('Época')
axes[1, 1].set_ylabel('Sparsity Coef')
axes[1, 1].set_yscale('log')
axes[1, 1].grid(True, alpha=0.3)

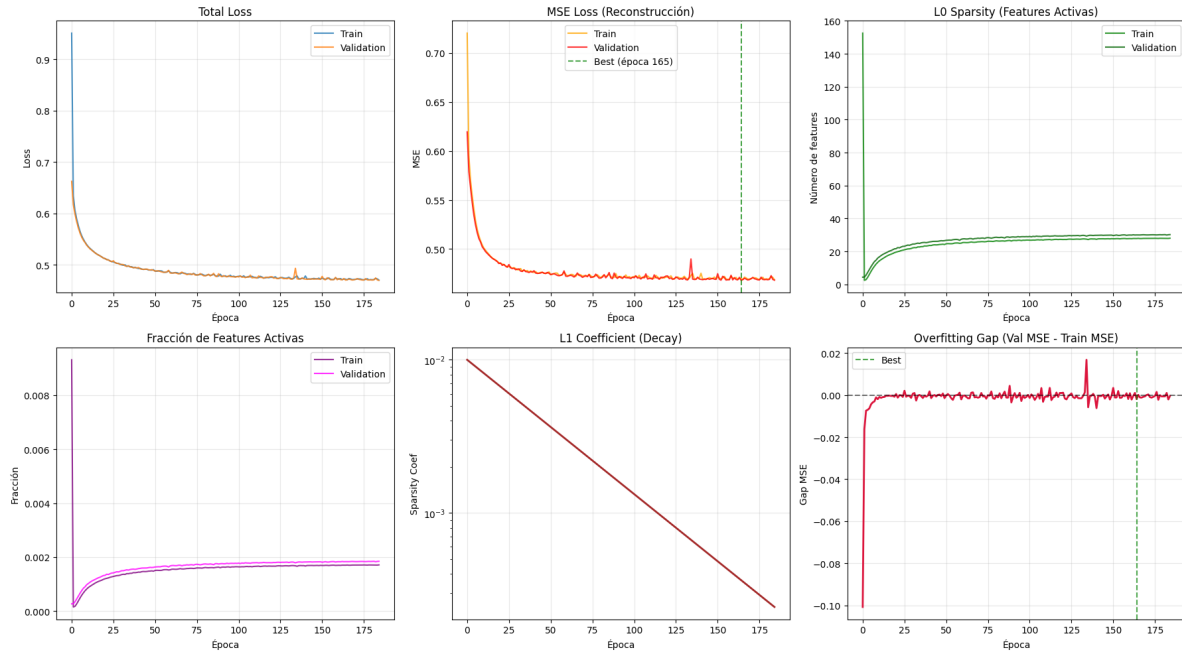
# Train vs Val MSE Gap
mse_gap = [v - t for v, t in zip(history['val_mse_loss'], history['train_mse_loss'])]
axes[1, 2].plot(mse_gap, color='crimson', linewidth=2)
axes[1, 2].axhline(0, color='black', linestyle='--', alpha=0.5)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[1, 2].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best Epoch')
axes[1, 2].set_title('Overfitting Gap (Val MSE - Train MSE)')
axes[1, 2].set_xlabel('Época')
axes[1, 2].set_ylabel('Gap MSE')
axes[1, 2].legend()
axes[1, 2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(CONFIG['plots_dir'] / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('  Gráficas guardadas exitosamente')

print('='*50)

```



=====
 Gráficas guardadas exitosamente
 =====

11. Resumen Final

```
# Resumen final completo
print('\n' + '='*70)
print(' RESUMEN FINAL DEL ENTRENAMIENTO')
print('='*70)

# Arquitectura
arch_summary = pd.DataFrame({
    'Componente': ['Input', 'Hidden', 'Parámetros'],
    'Valor': [
        f"{CONFIG['input_dim']}",
        f"{CONFIG['hidden_dim']} ({CONFIG['hidden_dim']/CONFIG['input_dim']:.1f}x)",
        f"{num_params:}"
    ]
})
```



```

print('\nArquitectura:')
print(arch_summary.to_string(index=False))

# Datos
data_summary = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ]
})
print('\nDatos:')
print(data_summary.to_string(index=False))

# Métricas finales
final_summary = pd.DataFrame({
    'Métrica': ['Train Loss', 'Val Loss', 'Train MSE', 'Val MSE', 'Train L0', 'Val L0'],
    'Valor': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['val_l0_sparsity'][-1]:.1f}"
    ]
})
print('\nMétricas Finales:')
print(final_summary.to_string(index=False))

# Archivos
files_summary = pd.DataFrame({
    'Tipo': ['Modelo', 'Gráficas'],
    'Ubicación': [
        str(checkpoint_path),
        str(CONFIG['plots_dir'] / 'training_curves.png')
    ]
})
print('\nArchivos Guardados:')
print(files_summary.to_string(index=False))

print('\n' + '='*70)

```

```
print('  ENTRENAMIENTO COMPLETADO EXITOSAMENTE')
print('='*70)
```

```
=====
RESUMEN FINAL DEL ENTRENAMIENTO
=====
```

Arquitectura:

Componente	Valor
Input	512
Hidden	16384 (32.0x)
Parámetros	16,794,112

Datos:

Conjunto	Muestras
Train	47,200
Validation	11,800
Total	59,000

Métricas Finales:

Métrica	Valor
Train Loss	0.4701
Val Loss	0.4696
Train MSE	0.4687
Val MSE	0.4682
Train L0	28.1
Val L0	30.2

Archivos Guardados:

Tipo	Modelo
decay_l6_1000g_final.pt	
Gráficas	c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1000games\ex5\plots\training_curves.png

```
=====
ENTRENAMIENTO COMPLETADO EXITOSAMENTE
=====
```

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
import subprocess

# Obtener directorio del experimento y nombre del PDF
exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_standard_l1-decay.ipynb'
output_path = exp_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
```

```
else:
    print(f' Error al generar PDF:')
    print(result.stderr)
```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1000games\ex5\sae_standard_l1-decay_l6_1000g_ex5_training.pdf

PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1000games\ex5\sae_standard_l1-decay_l6_1000g_ex5_training.pdf